Batti gli avversari (abbassatutto)

⇨ This task has a time limit of 10 minutes for submitting the solution. Once an input is requested, the timer will start automatically, and after it expires, it will no longer be possible to submit a solution for that input. *It is always possible to request a new input*, so don't worry if the timer expires: you will simply need to request and download a new input.

⇨ To help you with this task, we have prepared solution templates, which include only the parts for reading input and writing output (from keyboard and to screen). You can choose whether to read/write from files by uncommenting the appropriate lines of code.

You can download the templates from the main page for this problem.

Elio Muschio has decided to start playing FarmCraft. The game seems quite fun, but there's a small problem: Elio is still last in the global rankings, with zero points. Training to improve would be an option… but Elio has a much smarter plan in mind!

The FarmCraft leaderboard consists of N players, ordered from the highest score to the lowest. Each player in the i-th position has a score equal to P_i. In case of a tie in score, the player who reached that score first is ranked higher, so each position in the leaderboard is distinct.



In the game shop, there are N special boxes available, numbered from 1 to N, each with a cost of C_i coins. By purchasing the i-th box, Elio can reduce by 1 point the score of the player who at that moment occupies the i-th position in the leaderboard. Each box can be purchased any number of times. The leaderboard is updated after each score change, based on the players' new scores.

Since Elio Muschio is a sore loser, his plan is to use these boxes to lower the score of all other players to zero, so that everyone will have the same score as him.

Your task is to help Elio achieve his goal by spending the minimum possible number of coins.

Input Format

The first line of the input file contains an integer T, the number of test cases. Following this are T test cases, numbered from 1 to T. Each test case is preceded by a blank line.

Each test case is composed as follows:

- a line containing the integer N, the number of players on the leaderboard.
- a line containing the N integers P_1,...,P_N, the scores of the players on the leaderboard, from the first to the last.
- a line containing the N integers C_1,...,C_N, the cost of the boxes.

Output Format

The output file must contain the answer to the test cases you managed to solve. For each test case you solved, the output file must contain a line with the format "`Case #test:`", where test is the test case number (starting from 1), followed by the integer `ris`.

Constraints

- $T = 20$, in the input files you download there will be exactly 20 tes- cases.
- $1 \leq N \leq 10^5$.
- $0 \leq P_i \leq 10^5$.
- $0 \leq C_i \leq 10^5$.

Testcase Groups

The test cases are divided into 10 groups. Each group is worth 5 points which are obtained if all test cases in the group are solved correctly. Different groups may contain test cases with the same assumptions.

Constraints	Groups
C is sorted in ascending order	1, 2, 3
C is sorted in descending order	4, 5
$C_i \in \{0, 1\}$ for every i	6, 7
$N \leq 1000$	8
$\sum P_i \leq 100000$	9
No additional constraints	9, 10

Input/Output Examples

Input	Output
2 3 3 1 1 4 5 2 5 4 3 2 1 1 2 2 2 2 2	Case #1: 18 Case #2: 22

Explanation

In the **first sample case**, there are 3 players with initial scores $[3, 1, 1]$ and the costs of the boxes are $[4, 5, 2]$.

Remember that the leaderboard is updated **after each purchase**. Elio must reduce all scores to zero, choosing only one box each time. Here is a possible optimal sequence:

- Elio uses box 1 at a cost of 4 coins, the player in position 1 loses one point and drops to position 2: the leaderboard becomes $[2, 1, 1]$
- Elio uses box 1 at a cost of 4 coins: the player in position 1 loses one point and drops to position 2: the leaderboard becomes $[1, 1, 1]$
- Elio uses box 3 at a cost of 2 coins: the player in position 3 loses one point and remains in position 3: the leaderboard becomes $[1, 1, 0]$
- Elio uses box 1 at a cost of 4 coins: the player in position 1 loses one point and drops to position 2: the leaderboard becomes $[1, 0, 0]$
- Elio uses box 1 at a cost of 4 coins: the player in position 1 loses one point and drops to position 2: the leaderboard becomes $[0, 0, 0]$

Elio will therefore have used 4 times box number 1 and 1 time box number 3, spending a total of 18 coins, he could not have done better.

In the **second sample case**, Elio can use 11 times box number 1 paying 22 coins.